successful in achieving their 3:1 ratio. If not, if the queen really lives up to her name and the workers are her slaves and the obedient tenders of the royal nurseries, then we should expect the 1:1 ratio which the queen 'prefers' to prevail. Who wins in this special case of a battle of the generations? This is a matter that can be put to the test and that is what Trivers and Hare did, using a large number of species of ants.

The sex ratio that is of interest is the ratio of male to female reproductives. These are the large winged forms which emerge from the ants' nest in periodic bursts for mating flights, after which the young queens may try to found new colonies. It is these winged forms that have to be counted to obtain an estimate of the sex ratio. Now the male and female reproductives are, in many species, very unequal in size. This complicates things since, as we saw in the previous chapter, the Fisher calculations about optimal sex ratio strictly apply, not to *numbers* of males and females, but to *quantity of investment* in males and females. Trivers and Hare made allowance for this by weighing them. They took 20 species of ant and estimated the sex ratio in terms of investment in reproductives. They found a rather convincingly close fit to the 3:1 female to male ratio predicted by the theory that the workers are running the show for their own benefit.*

It seems then that in the ants studied, the conflict of interests is 'won' by the workers. This is not too surprising since worker bodies, being the guardians of the nurseries, have more power in practical terms than queen bodies. Genes trying to manipulate the world through queen bodies are outmanœuvred by genes manipulating the world through worker bodies. It is interesting to look around for some special circumstances in which we might expect queens to have more practical power than workers. Trivers and Hare realized that there was just such a circumstance which could be used as a critical test of the theory.

This arises from the fact that there are some species of ant that take slaves. The workers of a slave-making species either do no ordinary work at all or are rather bad at it. What they are good at is going on slaving raids. True warfare in which large rival armies fight to the death is known only in man and in social insects. In many species of ants the specialized caste of workers known as soldiers have formidable fighting jaws, and devote their time to fighting for the colony against other ant armies. Slaving raids are just a particular kind of war effort. The slavers mount an attack on a nest of ants belonging to a different species,